

Muhammed Rizwan

AI Engineer

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SUMMARY

AI Engineer with a Master's in AI Engineering and 1.5+ years of software engineering experience, now focused on building and deploying LLM and RAG systems. Shipped production-style applications using Python, LangChain, FastAPI, and Azure, with hands-on work in prompt engineering, hybrid retrieval, and model evaluation. Strong foundation in Python, deep learning (TensorFlow/Keras), and enterprise-grade debugging.

SKILLS

Languages & ML: Python, SQL, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

GenAI & RAG: LangChain, Prompt Engineering, LLMs (GPT-4, Llama 3), Embeddings, Vector Search, Hybrid Retrieval

Deployment & MLOps: FastAPI, Streamlit, Docker, Azure, Pinecone, FAISS, GitHub Actions, Git, Hugging Face Spaces, Linux

Evaluation: Precision, Recall, F1, MRR, Grad-CAM, A/B Testing, Model Explainability

EXPERIENCE

AI Engineer (Master's Thesis)

Jan 2025 – Jun 2025

Volvo Construction Equipment

Eskilstuna, Sweden

- Designed and deployed a RAG system on Azure Web Apps using GPT-4o Mini and Azure AI Search, improving stakeholder-rated accuracy from 5.88 to 6.42/10 across iterative releases.
- Built a data pipeline that scraped 24 customer success stories with BeautifulSoup and converted them into structured persona datasets (JSON) to augment the chatbot's knowledge base.
- Implemented hybrid retrieval (keyword + semantic + vector via HNSW) to improve response relevance for domain-specific queries over standalone vector search.
- Optimised prompting through Few-Shot vs CoT evaluation; CoT cut token usage by 41% and response time by 24%. Final system rated useful by 81.8% of stakeholders across R&D, marketing, and customer relations.

GenAI Intern

Aug 2024 – Sep 2024

PerformanceX

Remote

- Built a RAG-based sports coaching chatbot in 6 weeks using LangChain, Pinecone, and GPT-4, delivering two functional applications (data ingestion + chat interface).
- Benchmarked embedding and LLM models; selected all-MiniLM-L12-v2 (1.32s retrieval) and GPT-4 over slower local LLMs.
- Processed and structured data from 37 sources into a vector database with optimised chunking across 4 knowledge domains to improve retrieval relevance.

Software Engineer

Mar 2022 – Jun 2023

Allianz Technology

Trivandrum, India

- Maintained IBM FileNet P8 systems supporting enterprise clients across Germany and the UK, resolving 98% of incidents within SLA through root-cause analysis.
- Prevented production downtime with proactive Dynatrace monitoring and alerts across IBM WebSphere environments
- Ran system upgrades and environment configurations, gaining hands-on experience with enterprise deployment and debugging workflows.

PROJECTS

RAG Troubleshooting Assistant | Python, FastAPI, FAISS, BM25, Groq (Llama 3.3 70B), Docker

[Link](#)

- Built a hybrid retrieval RAG system (BM25 + FAISS + cross-encoder reranker) answering technical questions from 241 chunks of Docker docs with source citations. Evaluated on 10 test questions at 60% Recall@5, 0.75 MRR, 85% keyword coverage.
- Added production features (retry logic, rate limiting, response caching, token tracking, timeouts). Containerised with Docker, deployed on Hugging Face Spaces, and set up a CI-gated GitHub Actions pipeline that blocks merges if retrieval quality drops below thresholds.

AI Workflow for Job Search | Python, Ollama

[GitHub](#)

- Built a daily pipeline that pulls AI jobs from the Swedish Employment Service API, filters irrelevant listings, and uses an LLM to score the rest with match percentage and reasoning; runs nightly via Task Scheduler, processing ~25 jobs/day and emailing the best matches to Gmail.

EDUCATION

Jönköping University | Master of Science in AI Engineering, Sweden

2023 – 2025

Mangalore Institute of Technology | Bachelor of Engineering in Computer Science, India

2017 – 2021